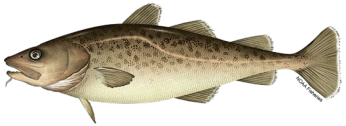
From heatwaves to recovery:

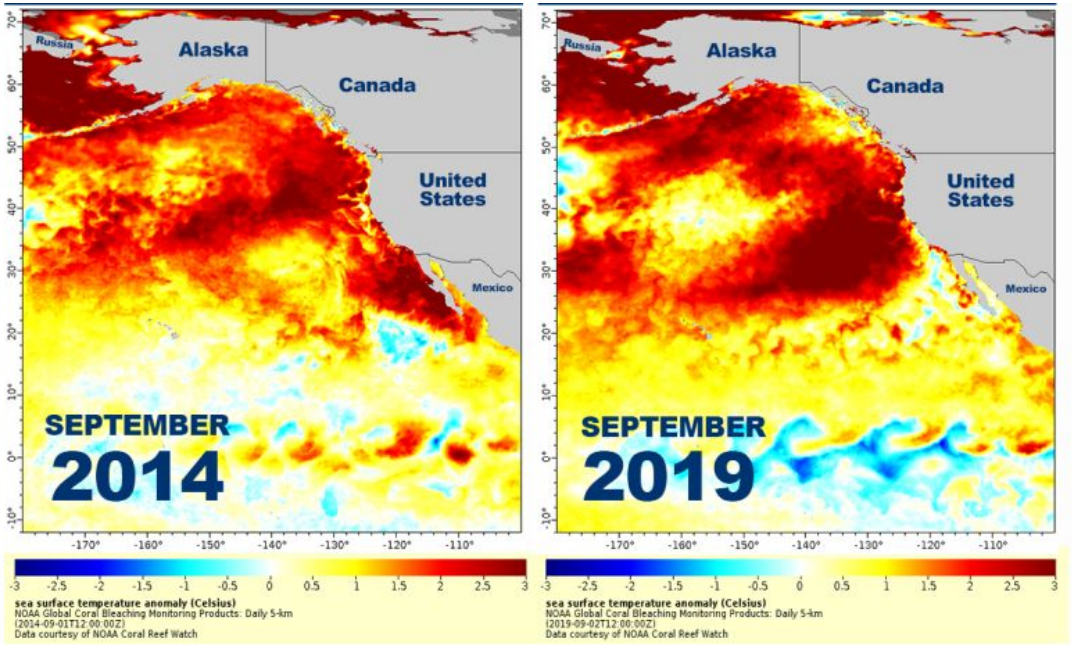
Temperature-linked performance and resilience in Gulf of Alaska Pacific cod



Alaska Marine Science Symposium, Anchorage, AK 2026

Laura Spencer, Univ. of Washington SAFS & NOAA AFSC
Ben Laurel, NOAA AFSC Newport, OR
Louise Copeman, NOAA AFSC Newport, OR
Mary Beth Rew-Hicks, NOAA AFSC Newport, OR
Ingrid Spies, NOAA AFSC Seattle, WA
Steven Roberts, University of Washington SAFS

Heatwaves!



[NOAA feature story](), *September 05, 2019*



Pacific cod, *Gadus macrocephalus*

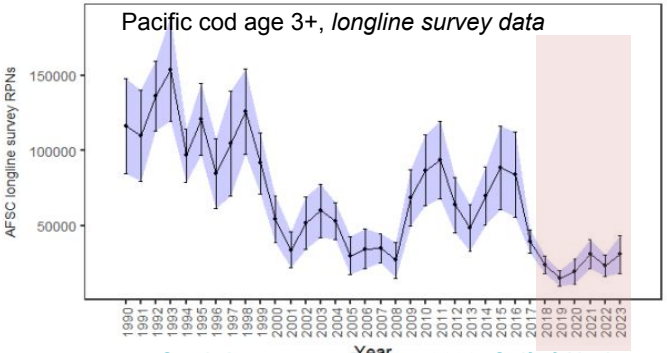
Extremely low cod numbers lead feds to close the Gulf of Alaska fishery for the first time

Alaska Public Media
Published December 6, 2019 at 8:22 PM AKST



Pacific cod (Photo Courtesy of Holland Dotts & the Alaska Marine Conservation Council)

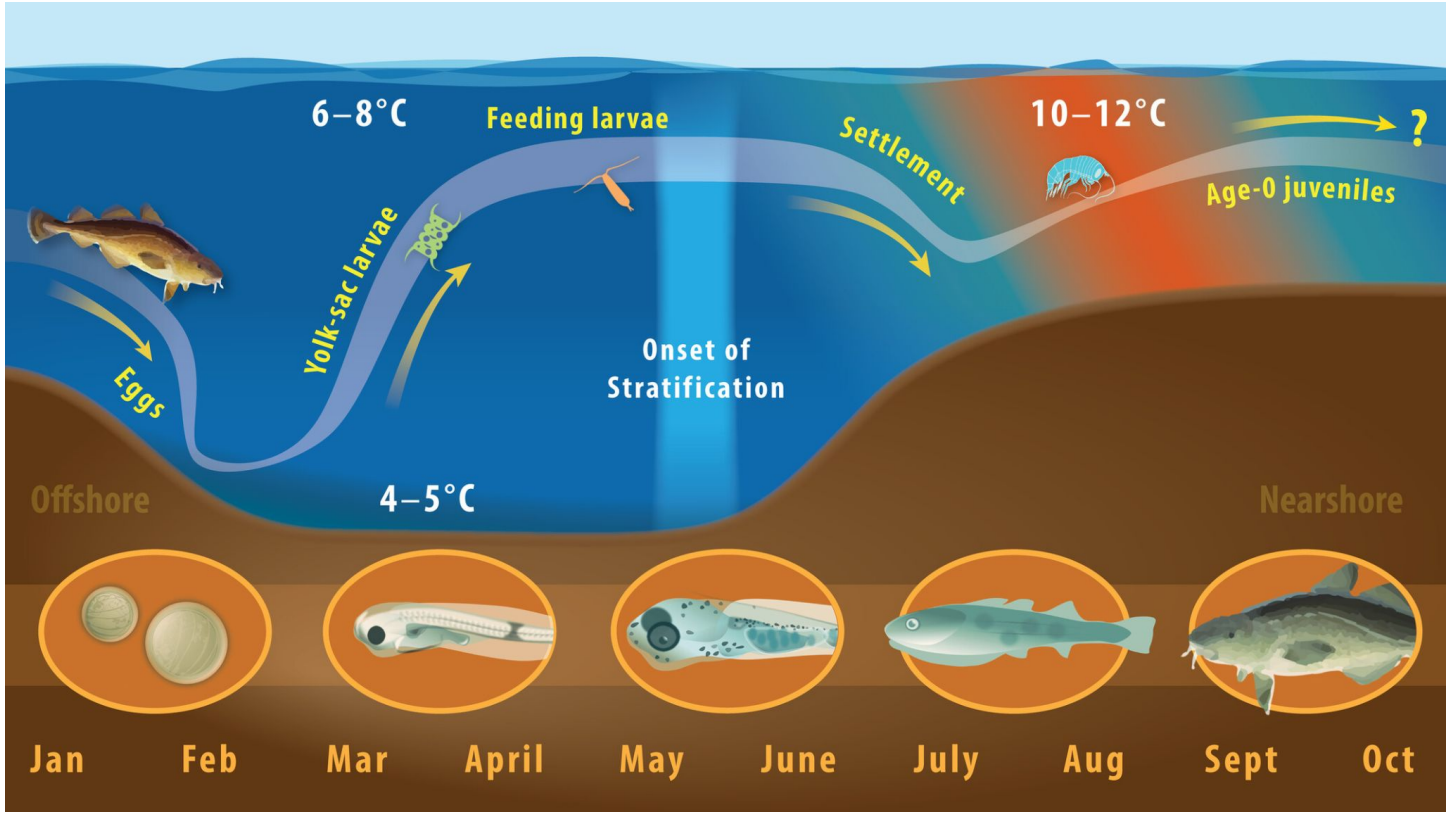
[Alaska Public Media](), *December 06, 2019*



[Stock Assessment Report, 2023, Gulf of Alaska](),



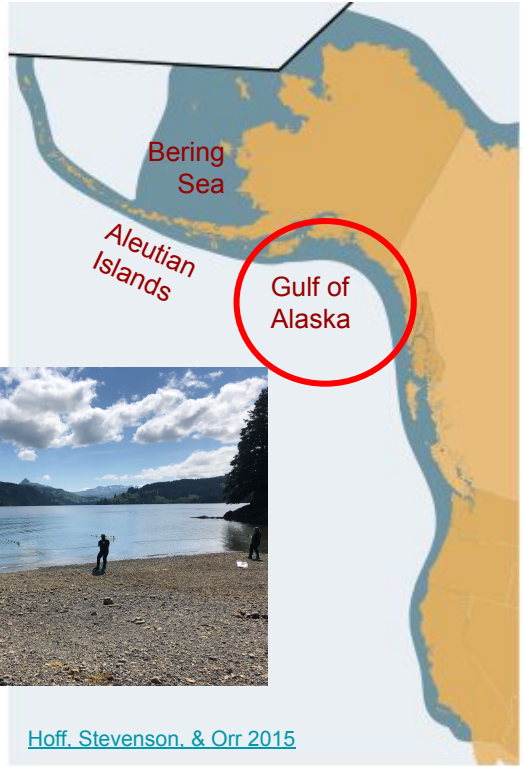
Pacific cod early life stages are very sensitive to temperature



Laurel et al. 2023, *in Fish & Fisheries*



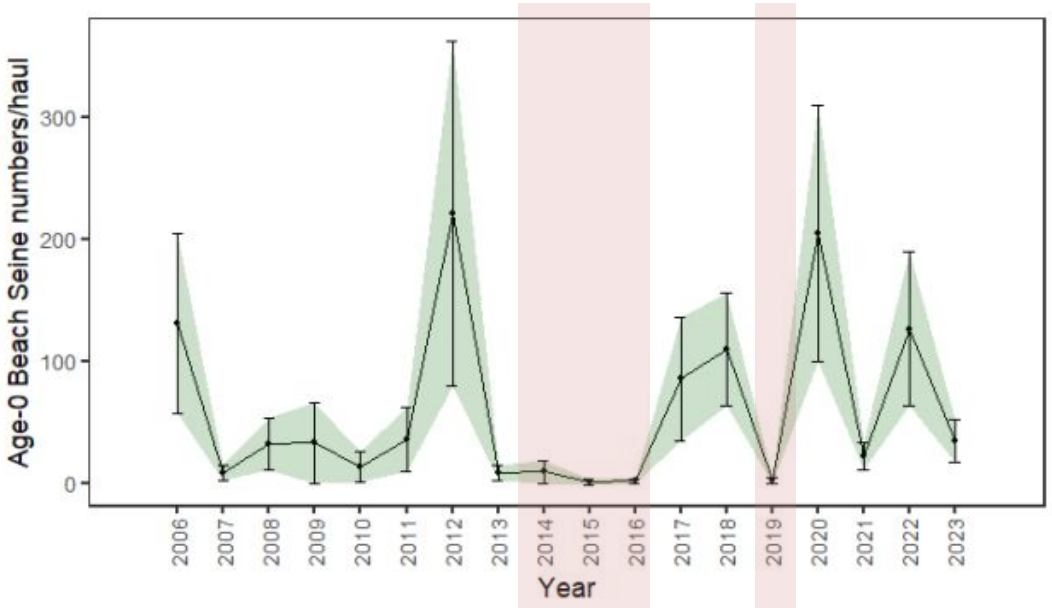
Low abundances likely related to low recruitment during heat waves





Hoff, Stevenson, & Orr 2015

Pacific cod juvenile (age-0) *beach seine survey CPUE*



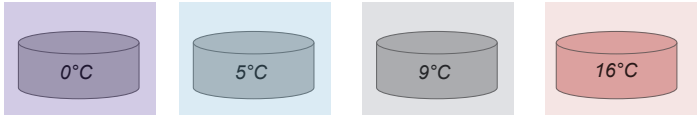
Stock Assessment Report, 2023, Gulf of Alaska,



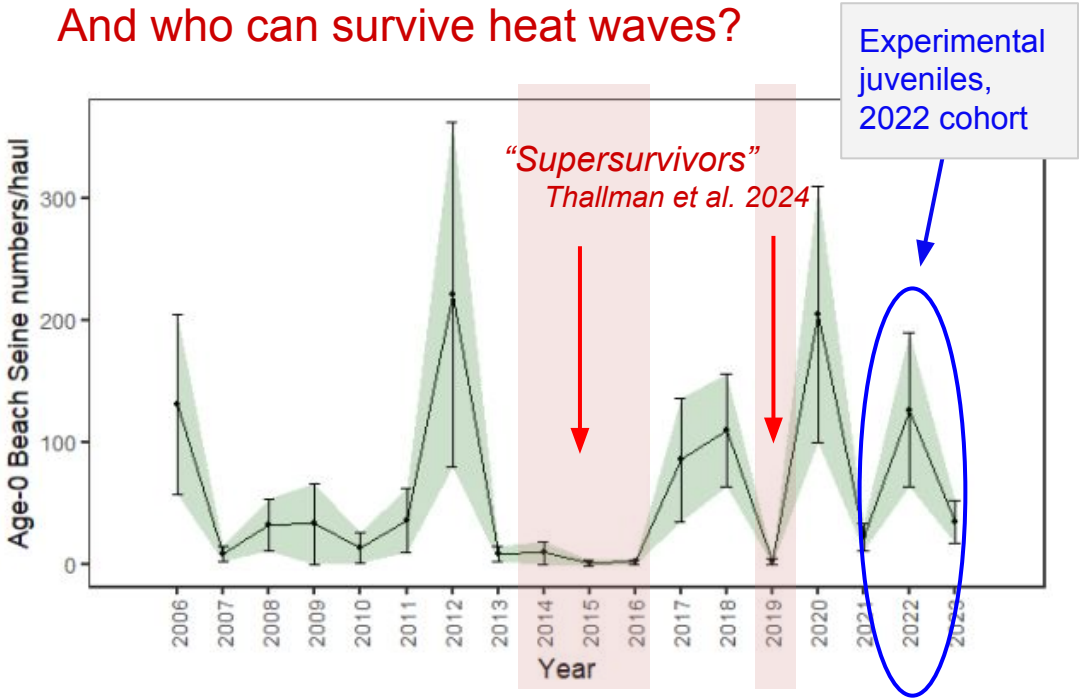
Question: How does temperature affect juvenile Pacific cod?

Juvenile Temperature Experiment





And who can survive heat waves?



Stock Assessment Report, 2023, Gulf of Alaska,



“Genome to Phenome” dataset across temperatures for Pacific cod





→ Individuals were tagged ←

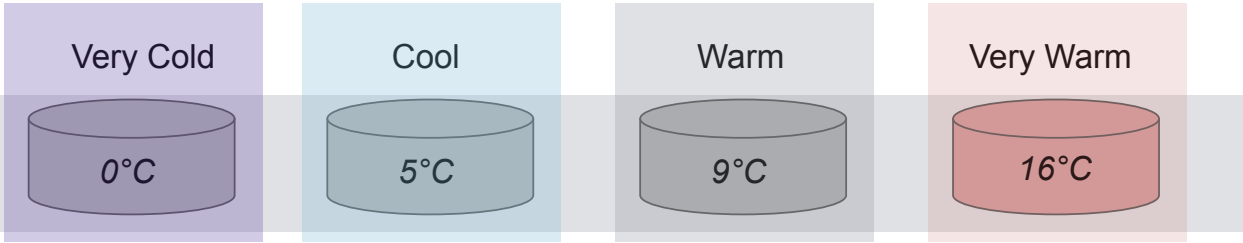
a. Genetics (lcWGS)

b. Gene expression in liver, n=18/temp (RNASeq)

c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival



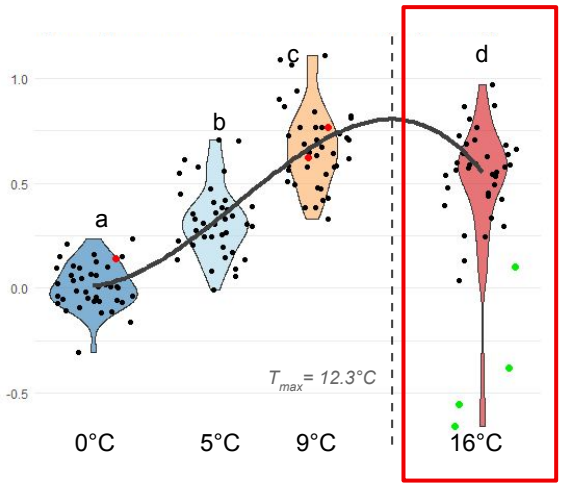
n=40 / temp.
4 week treatment



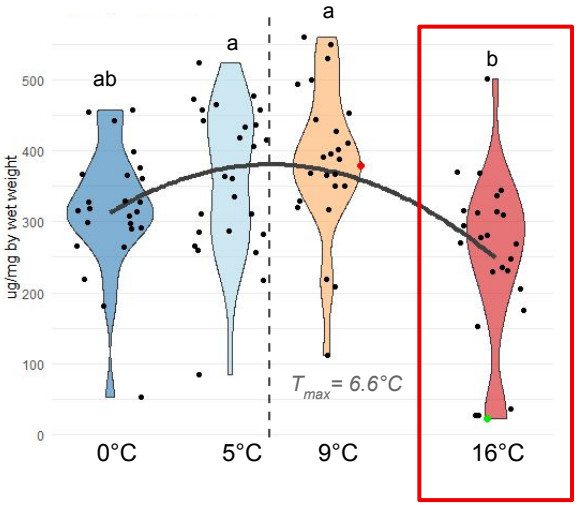


Warming slows growth, reduces energy reserves (lipid), & increases mortality - likely to impact overwinter survival.

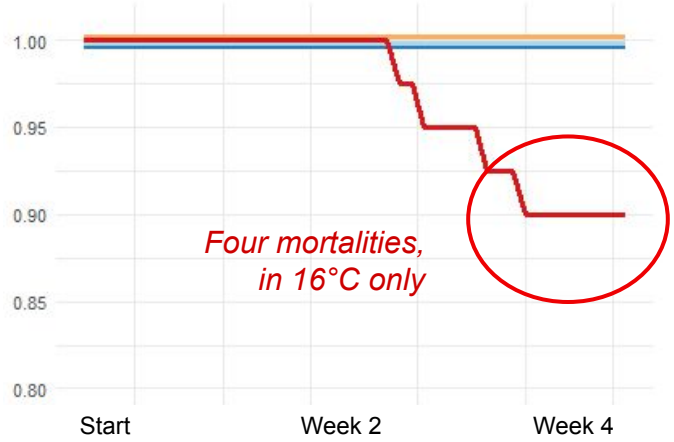
Growth Rate (weight, %/day)



Lipid Content in liver (mg)



Survival

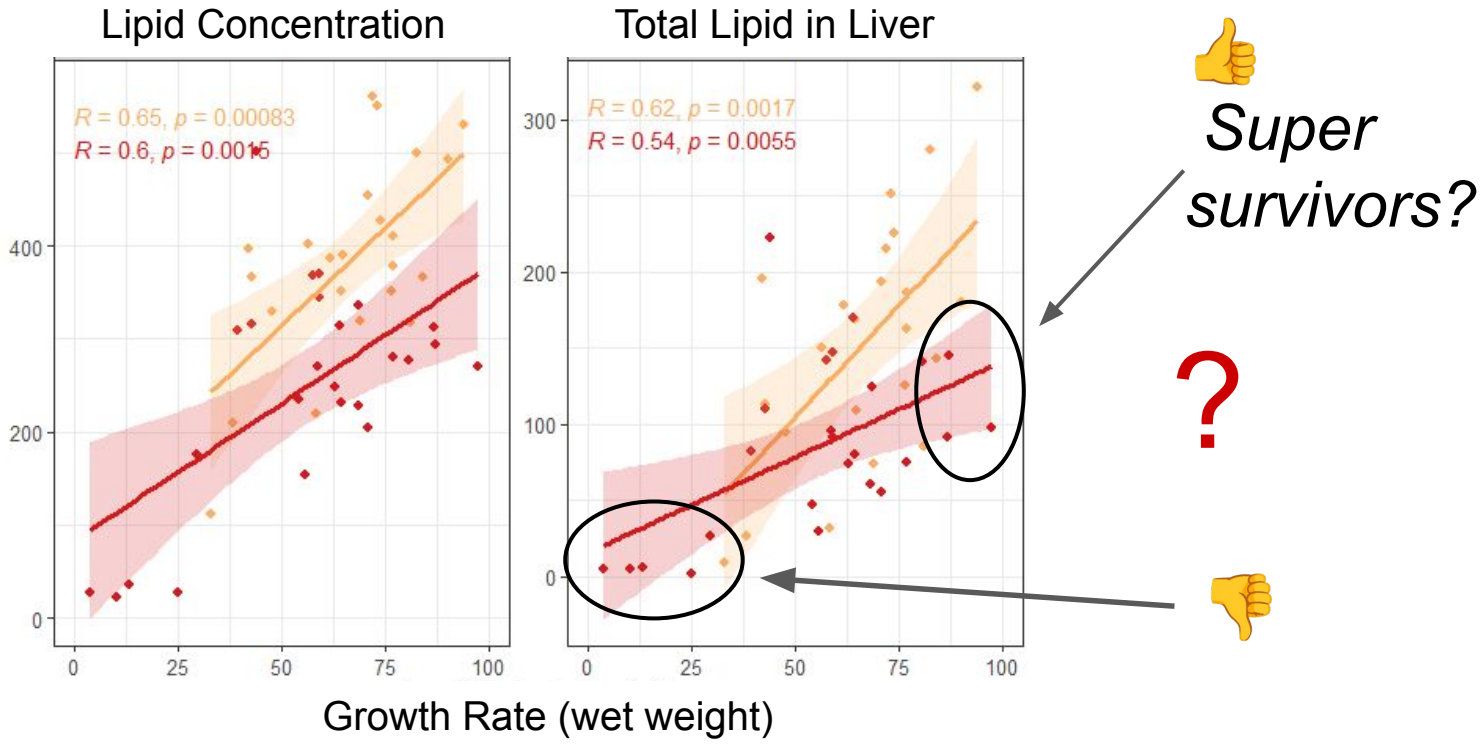


n=40/temperature

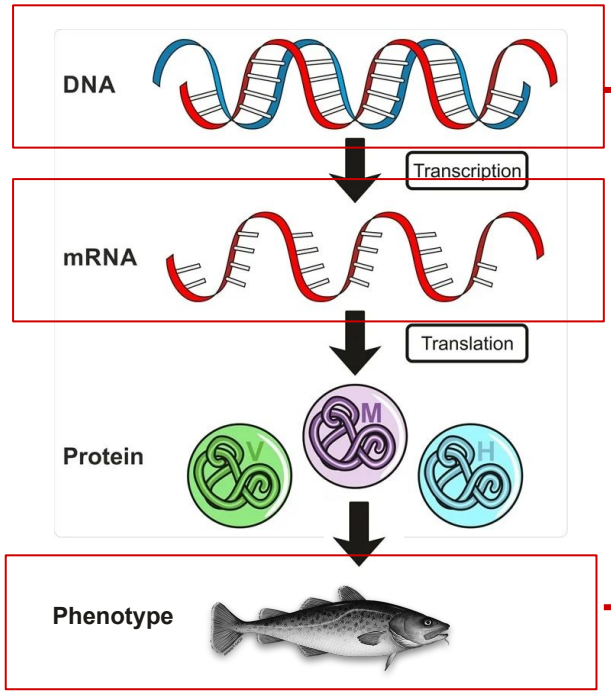
In press, GCB-Comm



The fastest growers in warm temperatures also have the most energy reserves (lipid) in liver



“Genome to Phenome” dataset across temperatures



Adapted from: udaix/Shutterstock.com

→ Individuals were tagged ←

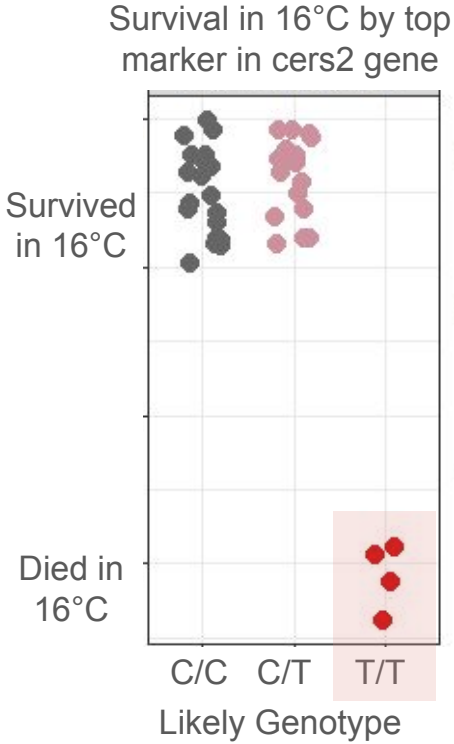
a. Genetics (lcWGS)
b. Gene expression in liver (RNASeq)
c. Fitness Traits
   - Growth rate
   - Liver & Body Condition
   - Liver lipid concentration, content
   - Survival

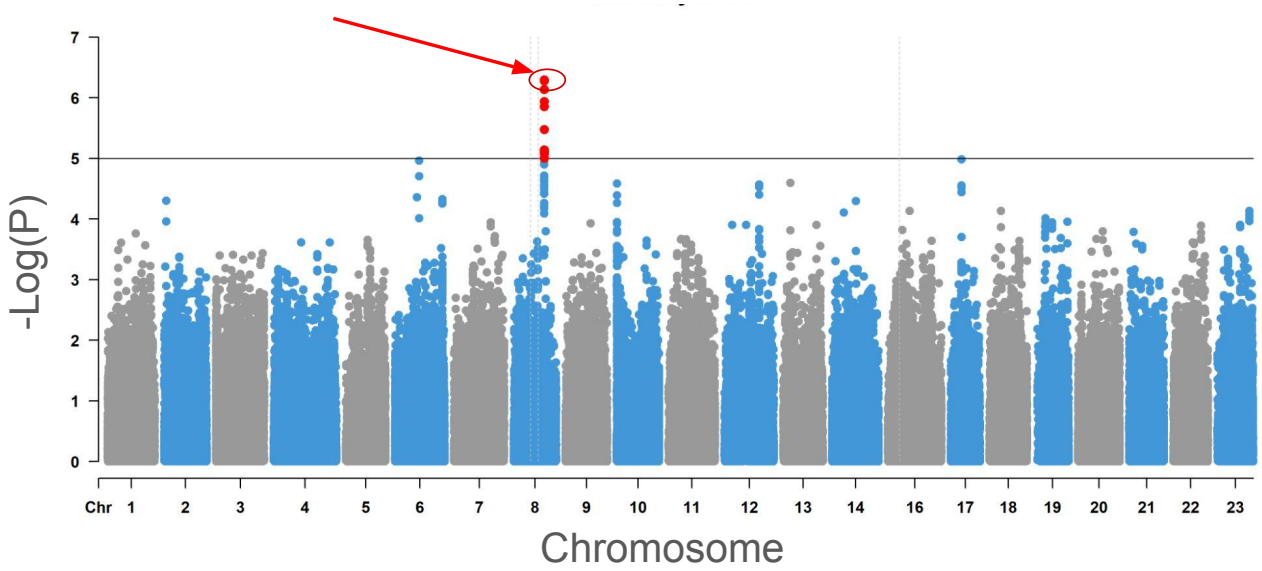
Which **sites on genome** are associated with **fitness traits?**

Genome-Wide Association Studies (GWAS) using single nucleotide polymorphisms (SNPs)



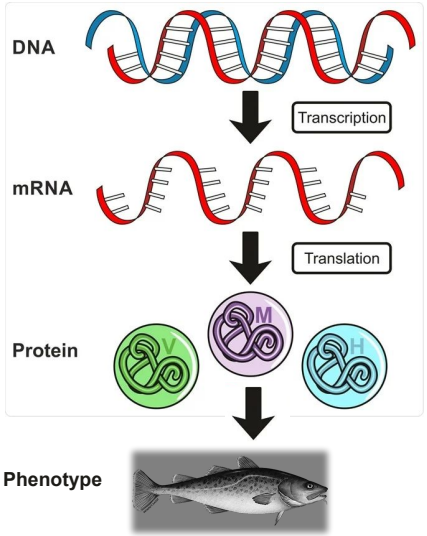
SNP markers in 1 gene, ceramide synthase 2 (cers2) associated with **mortality in warming (16°C)** in Pacific cod juveniles





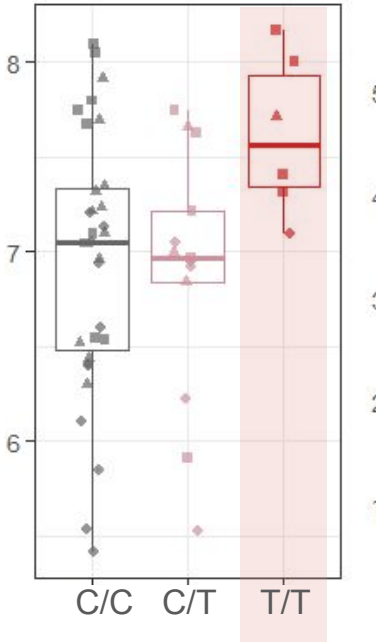


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates



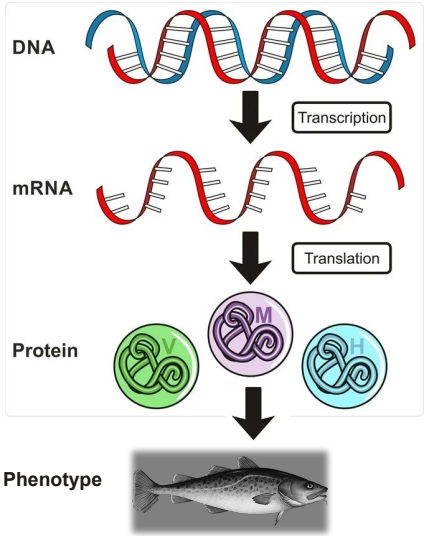
Adapted from: udaix/Shutterstock.com

Ceramide synthase 2 expression in liver



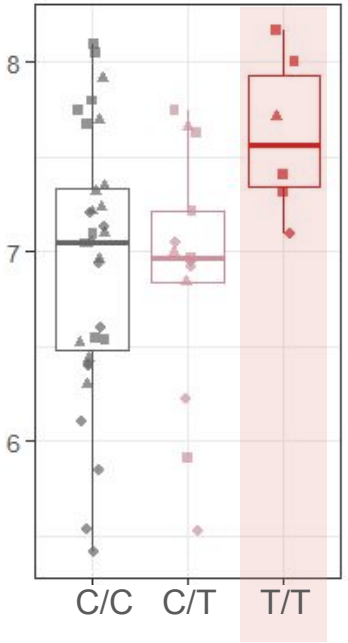


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

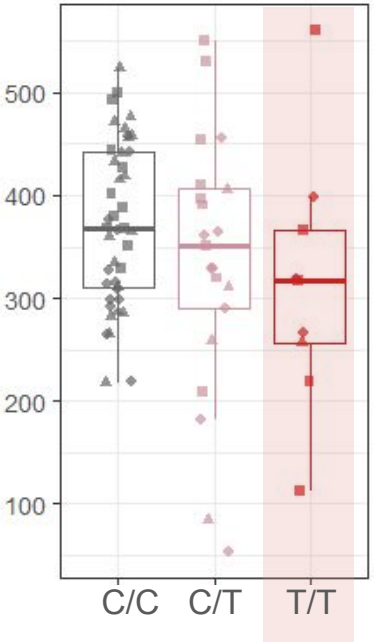


Adapted from: udaix/Shutterstock.com

Ceramide synthase 2 expression in liver

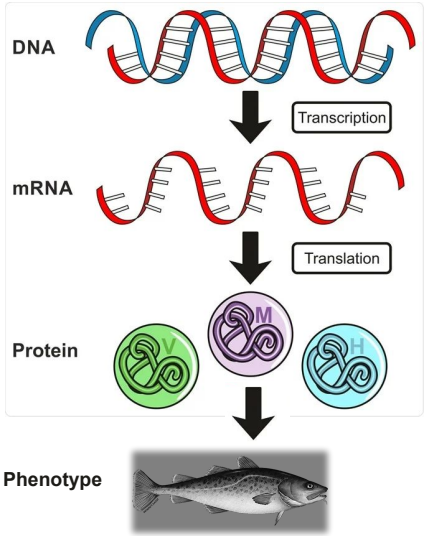


Lipid concentration in liver



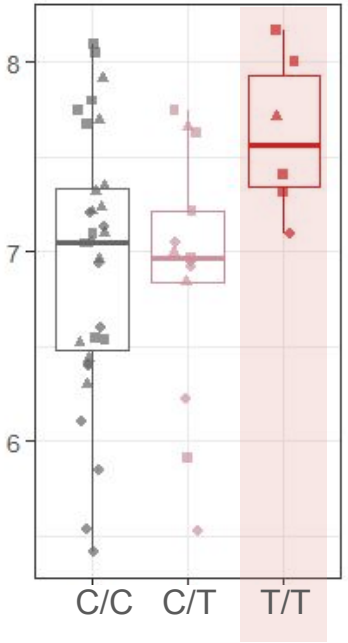


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

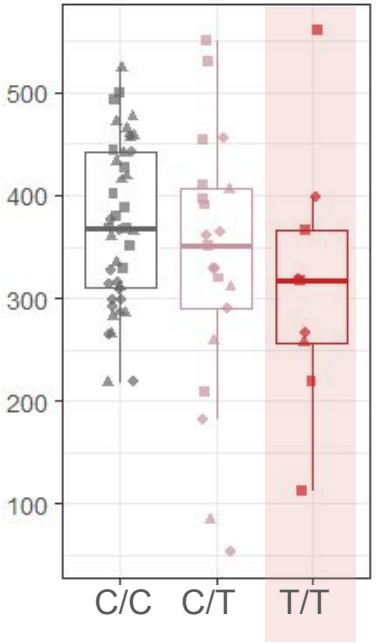


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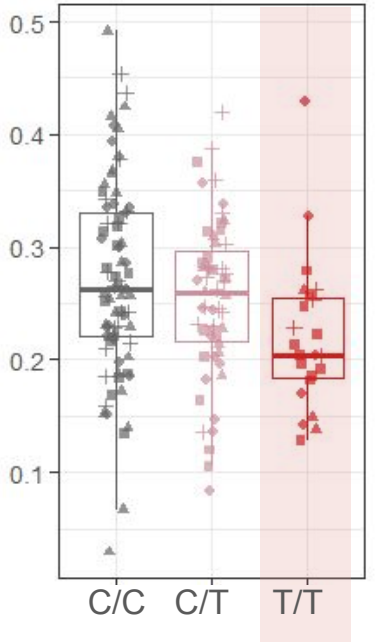
Ceramide synthase 2 expression in liver



Lipid concentration in liver



Growth rate during acclimation (all at 6°C)





Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

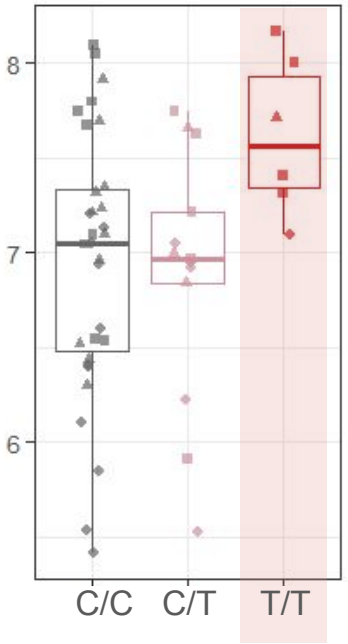
Cers2 synthesizes long-chain fatty acids (ceramides), involved in membrane structure, stress signaling

Who can survive heat waves?

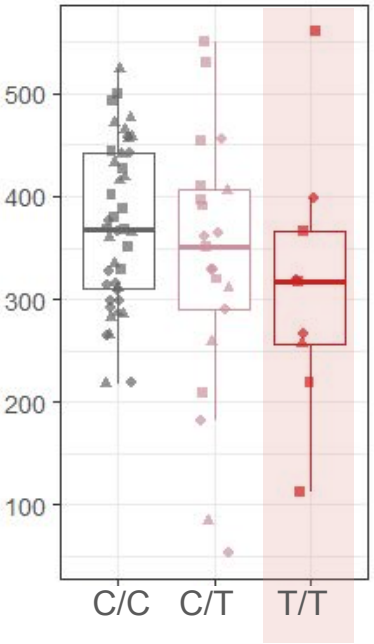
Maybe not young cod with T/T in cers2 marker

… *not necessarily causal, and traits are polygenic!!*

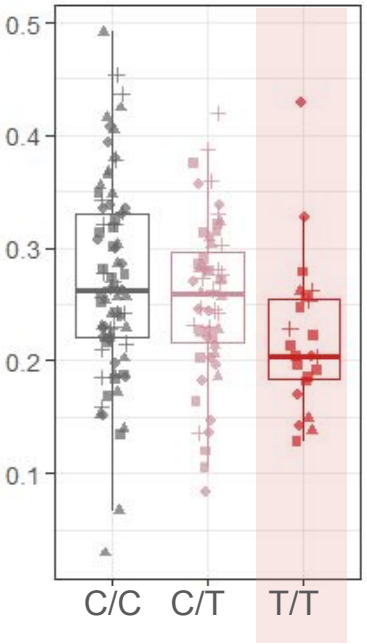
Ceramide synthase 2 expression in liver



Lipid concentration in liver

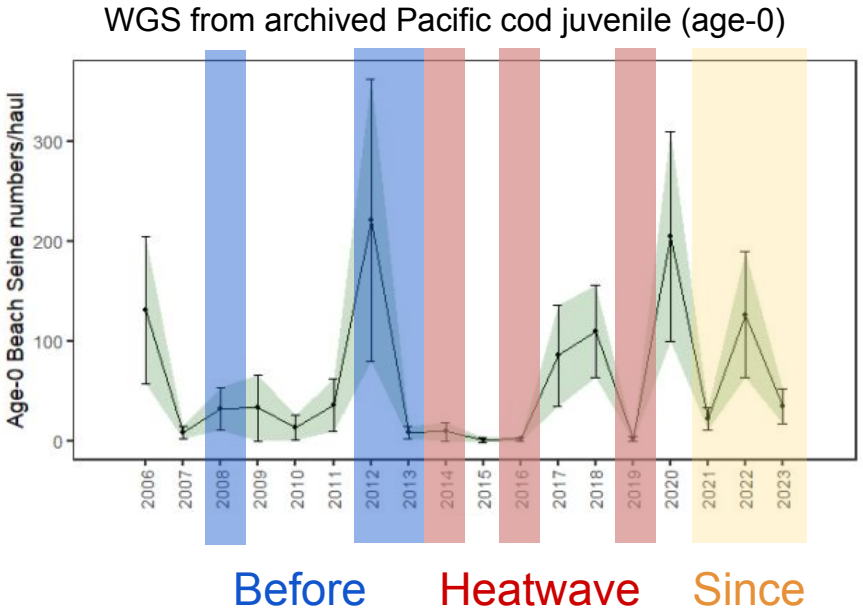


Growth rate during acclimation (all at 6°C)





Who actually survived heat waves? TBD!



Genetic Time Series

- Juveniles from nine distinct years, spanning heatwaves (2008-2023)
- ~40 per year, all in July off Kodiak
- Analysis: track allele frequencies at candidate markers over heatwaves
- **Signals of selection for trait-associated markers?**



Pacific cod juvenile genome-to-phenome temperature study

Experimental results:

- Juvenile cod growth & energy allocation is highly sensitive to temperature.
- **Low recruitment in GOA during heat waves likely related to high mortality rates through juvenile stage,** and **low overwintering survival due to fewer lipid reserves.**

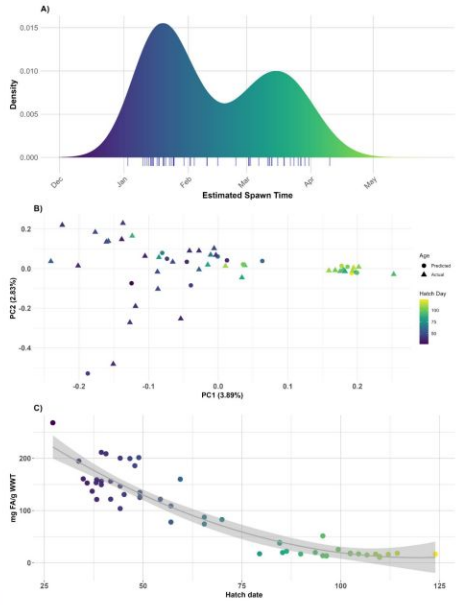
Applications:

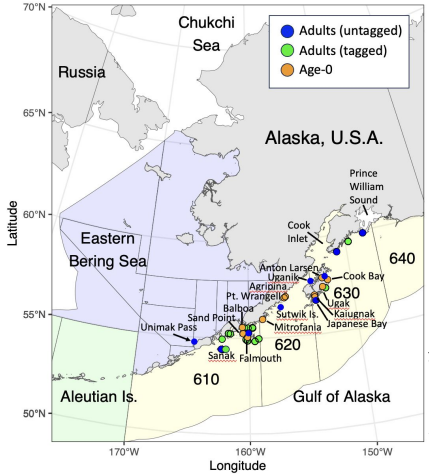
- Stock assessments (AI, GOA) recently **adjusted natural mortality rates** to account for temperature
- Molecular resources for Pacific cod
  - SNP Reference Panel for Pacific cod in GOA / EBS - *for imputation, etc.*
  - Genes that are temperature sensitive in early stages - *liver transcriptomes*
  - Markers of fitness traits - *monitor populations*

So, what's the outlook for GOA cod?

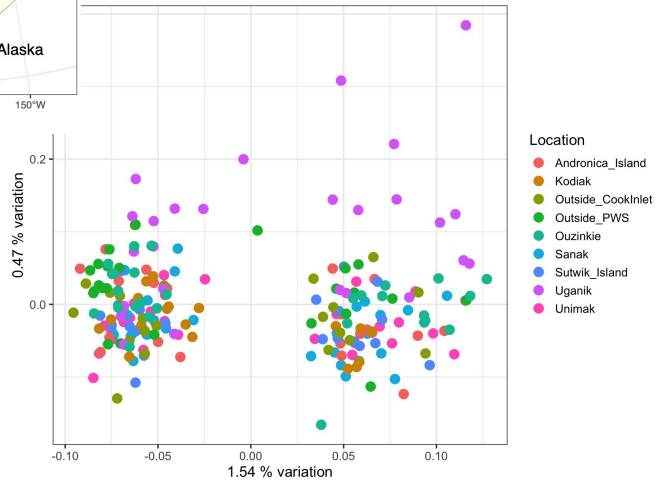


There is variation within wGOA/EBS Pacific cod, which is good!





New research! *There are two genetically distinct groups in GOA with varying depth (and possibly temperature) preferences - "ecotypes"*



First evidence of genetically distinct winter- and spring-spawning components of Pacific cod in the Gulf of Alaska

Authors: Benjamin Laurel, Jessica Miller, Louise Copeman, Alisa Abookire, Laura Spencer, Sara Schaal, Mary Beth Rew Hicks, Nicholas Strait, Lauren Rogers, and Ingrid Spies | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 16 January 2026 • https://doi-org.offcampus.lib.washington.edu/10.1139/cjfas-2025-0168

Contributors & Funders

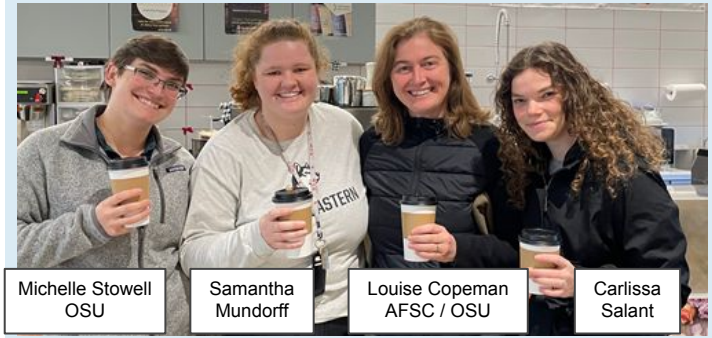
Juvenile Collection & Experimental Data


Ben Laurel, AFSC


Mary Beth Rew Hicks, AFSC


Paul Iseri, AFSC

Lipid Data


Michelle Stowell OSU

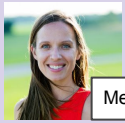
Samantha Mundorff

Louise Copeman AFSC / OSU

Carlissa Salant



‘Omics Data & Bioinformatics


Me


Ingrid Spies, AFSC


Sara Schaal, AFSC & UO


Wes Larson, AFSC


Steven Roberts, UW


Laura Timm, AFSC


Kathleen Durkin, UW